

Assignment 1

IB3702 Mathematics for Machine learning

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Month Day, Year

Problem 1

Question 1.1

- (a) Considering a_0 is 1, we have to simply fill in 1 where it says a_n in the formula. The result of this will be a_1 . Then we will take a_1 and fill it in the formula to get a_2 . We will repeat this process until we reach a_4 .

- $a_0 = 1$
- $a_1 = \frac{1}{2}(1 + \frac{2}{1}) = \frac{1}{2} * 3 = \frac{3}{2}$ So $a_1 = \frac{3}{2}$
- $a_2 = \frac{1}{2}(\frac{3}{2} + \frac{2}{\frac{3}{2}}) = \frac{1}{2} * (\frac{3}{2} + \frac{4}{3}) = \frac{1}{2} * \frac{17}{6} = \frac{17}{12}$ So $a_2 = \frac{17}{12}$
- $a_3 = \frac{1}{2}(\frac{17}{12} + \frac{2}{\frac{17}{12}}) = \frac{1}{2} * (\frac{17}{12} + \frac{24}{17}) = \frac{1}{2} * \frac{577}{204} = \frac{577}{408}$ So $a_3 = \frac{577}{408}$
- $a_4 = \frac{1}{2}(\frac{577}{408} + \frac{2}{\frac{577}{408}}) = \frac{1}{2} * (\frac{577}{408} + \frac{816}{577}) = \frac{1}{2} * \frac{665,857}{235,416} = \frac{665,857}{470,832}$ So $a_4 = \frac{665,857}{470,832}$

- (b) answer/solution to this problem

Problem 2

- (a) answer/solution to this problem
(b) answer/solution to this problem

Problem 3

- (a) answer/solution to this problem
(b) answer/solution to this problem

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